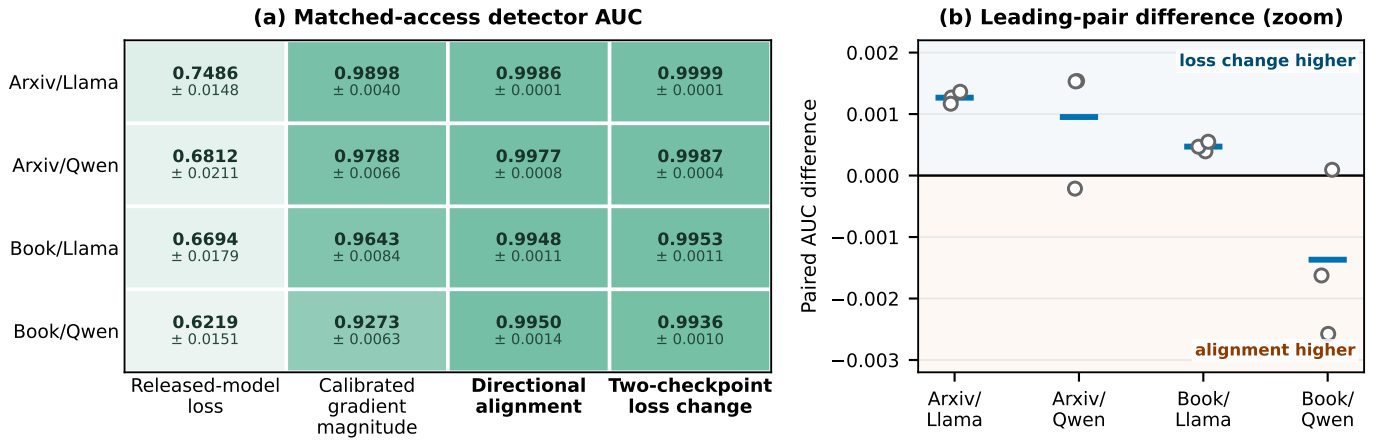
(a) Matched-access detector AUC
Arxiv/Llama
Arxiv/Qwen
Book/Llama
Book/Qwen
0.7486 ± 0.0148
0.9898 ± 0.0040
0.9986 ± 0.0001
0.9999 ± 0.0001
0.6812 ± 0.0211
0.9788 ± 0.0066
0.9977 ± 0.0008
0.9987 ± 0.0004
0.6694 ± 0.0179
0.9643 ± 0.0084
0.9948 ± 0.0011
0.9953 ± 0.0011
0.6219 ± 0.0151
0.9273 ± 0.0063
0.9950 ± 0.0014
0.9936 ± 0.0010
Released-model loss
Calibrated gradient magnitude
Directional alignment
Two-checkpoint loss change
(b) Leading-pair difference (zoom)
Paired AUC difference
0.002
0.001
0.000
−0.001
−0.002
−0.003
loss change higher
alignment higher
Arxiv/ Llama
Arxiv/ Qwen
Book/ Llama
Book/ Qwen